

Analysis and Compensation for the Impact of IOC Half-Wave Voltage Thermal Dependence Errors on Scale Factor of IFOGs

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Abstract—Interferometric fiber optical gyroscopes (IFOGs) serve as rotation rate sensors operating on the Sagnac effect. The digital phase ramp feedback detection scheme is currently the global industry standard. However, the half-wave voltage of the integrated optical chip (IOC) used for phase modulation demonstrates high sensitivity to ambient temperature fluctuations, significantly impacting the IFOG scale factor. This study investigates the thermal dependence of the IOC half-wave voltage and its effect on scale factor accuracy. We further analyze how reset errors during digital phase ramp resets, induced by half-wave voltage variations, influence scale factor performance, establishing a quantitative relationship between reset error and scale factor inaccuracy. The principles of a secondary closed-loop system for tracking half-wave voltage fluctuations are presented, along with its operational limitations. Leveraging the thermal dependence model of a lithium niobate modulator's half-wave voltage, we propose a cost-effective half-wave voltage tracking method for scale factor compensation. Verification results demonstrate tracking accuracy exceeding 406 ppm.

Index Terms—Compensation method, fiber optical gyroscope, integrated optical chip, reset error, scale factor.

I. INTRODUCTION

INTERFEROMETRIC Fiber optical gyroscopes (IFOGs) are rotation rate sensors that operate based on the Sagnac effect. They offer several advantages, including an all-solid-state design, compact size, high precision, and strong adaptability to various environmental conditions [1], [2]. As a result, IFOGs have found widespread applications in aviation, aerospace, defense, and energy sectors. Currently, IFOG-based Attitude and Heading Reference Systems (AHRS) developed by KVH in the United States are extensively used in aircraft such as

the Embraer 145 regional jet, Boeing 777, and Cessna Excel business jet. Honeywell's LASEREF VI Micro Inertial Reference System, based on IFOGs, is installed on Airbus A320 series and A330/A340 aircraft. Emcore's HawkeyeTM EG IFOG series is widely used for navigation, guidance, and control in autonomous drones and fixed-wing aircraft. iXBlue's IFOGs have been deployed in submarines [3], [4], [5], [6]. In 2011, IFOGs were applied to the Chinese space station Tiangong-1, setting a precedent for their use in space stations. In 2017, Photonic Crystal Fiber optical gyroscopes (PCFOGs) were launched into space aboard the Tianzhou-1 cargo spacecraft, marking an international milestone in the space applications of PCFOGs [7], [8], [9], [10]. As the development of low-cost and miniaturized IFOGs continues, they are expected to demonstrate strong competitiveness in low-altitude flight and autonomous driving fields in the future [11].

Similar to other high-precision sensors, IFOGs are sensitive to external environmental uncertainties, which cause discrepancies between output values and true values. Extensive research into IFOG error mechanisms has been conducted over many years, with researchers and engineers worldwide developing significant suppression and compensation techniques. Temperature remains the primary environmental factor affecting IFOGs performance, influencing nearly all optical and electrical components within IFOGs systems [12], [13], [14]. Modern IFOGs employ digital closed-loop detection schemes to extend the measurement range and improve scale factor linearity. In this architecture, the integrated optical chip (IOC) generates the closed-loop negative feedback modulation phase [15]. Crucially, the feedback channel gain exhibits high sensitivity to ambient temperature fluctuations due to the temperature dependence of the voltage required for IOC phase modulation. Since the IFOG scale factor is determined by this gain, effective compensation for IOC temperature sensitivity is essential. The digital closed-loop system modulates the IOC using step signals for negative feedback control. However, these signals cannot increase or decrease indefinitely—they reset when the modulation phase reaches 2π . Deviations between the applied and ideal modulation voltages cause abrupt signal transitions at photodetectors during reset events [16], [17]. These transitions persist even when the primary loop control is stable. Importantly, the demodulation process introduces errors into angular velocity calculations, becoming significant factors that limit IFOGs performance.

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Moreover, half-wave voltage errors and reset errors always coexist simultaneously. By adjusting the half-wave voltage to reduce the monitored reset error to zero, the thermal dependence error of the IOC can be reduced, thus giving rise to the secondary feedback loop technique. While this approach effectively reduces the impact of half-wave voltage errors in most cases, it is not always perfect. This method involves continuously adjusting the feedback gain by comparing demodulation values from the two modulation half-cycles before and after reset events. Through adjustments to the IOC's drive voltage, the reset error is driven to approach zero, thereby ensuring the actual modulation phase closely aligns with the ideal phase [18]. However, this method is susceptible to failure under two conditions: The first involves high dynamic conditions where frequent step signal resets occur, the reset error extraction process itself can introduce significant inaccuracies, increasing error risk. The second occurs during low rotation rate inputs, where excessively long reset intervals may cause secondary feedback loop failure. The fundamental reason is that the secondary feedback loop must operate within a relatively stable range. Furthermore, the implementation complexity of the secondary feedback loop requires substantial hardware circuitry or sophisticated software algorithm support. We still need a cost-effective solution.

In addition, the use of four-state modulation is a common method to address both the temperature stability issues of lithium niobate devices and the drift caused by electronic circuits in analog demodulation. However, the application of four-state modulation has certain limitations, as it is significantly influenced by spectral asymmetry of the light source. This modulation technique is often employed in high-precision FOGs [1]. A distinguishing feature of high-precision FOGs in terms of optical design is the use of a broadband superfluorescent erbium-doped fiber source. This type of source provides a stable output spectrum and high output power, achieved through amplified spontaneous emission from Er^{3+} ions in the doped fiber under pump light excitation. The resulting spectrum exhibits an asymmetric profile with a leading peak and a trailing peak. To produce a well-symmetrized output spectrum, an optical filter is required to remove unwanted spectral components. Therefore, a prerequisite for applying the four-state modulation method is ensuring the symmetry of the light source's output spectrum. This requirement necessitates the use of additional optical components and the implementation of complex correction algorithms.

This paper analyses the temperature dependence of the half-wave voltage in IOCs for IFOGs and investigates how ambient temperature fluctuations, which alter the IOC half-wave voltage, impact the scale factor. It further derives the quantitative influence of reset errors on the scale factor. The working principle of the secondary feedback loop for compensating IOC half-wave voltage variations is presented, along with an analysis of its application limitations. Building upon the stable linear relationship between IOC half-wave voltage and temperature, we propose a cost-effective half-wave voltage tracking method to compensate for scale factor errors induced by changes in half-wave voltage. This approach offers significant practical value, particularly for medium-to-low precision gyroscopes or in high-dynamic measurement scenarios.

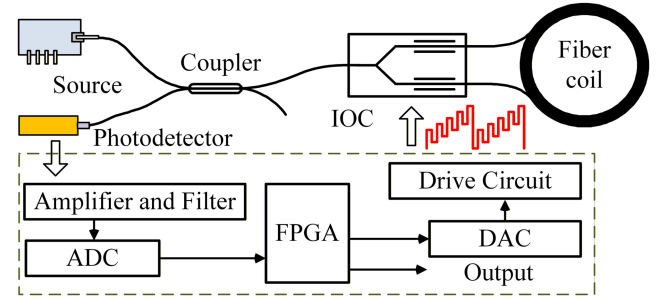


Fig. 1. Structure diagram of a single-axis IFOG.

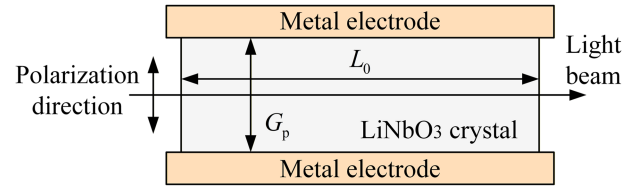


Fig. 2. The structure of the IOC's phase modulation section.

II. IMPACT OF HALF-WAVE VOLTAGE THERMAL DEPENDENCE ERRORS ON SCALE FACTOR

A. Scale Factor Error Caused By Half-Wave Voltage Error

The structure diagram of a single-axis IFOG is shown in Fig. 1 and comprises two primary components: optical elements and circuitry. The optical elements include a light source, coupler, IOC, optical fiber coil, and photodetector, which together form a Sagnac interferometer. The circuitry is designed to implement a phase-sensitive detection method based on digital techniques. The interference signal produced by the Sagnac interferometer is captured by the photodetector and subsequently processed through filtering and demodulation. Feedback signals are generated within a Field Programmable Gate Array (FPGA) and converted to voltage levels using Digital-to-Analog Converters (DACs), which drive the IOCs to adjust the phase of the beams accordingly.

The IOC is a crucial component in interferometric digital closed-loop IFOGs. It performs the functions of polarizing, splitting, and recombining the light beam, with its most critical role being phase modulation. At present, the vast majority of IOCs used in interferometric IFOGs employ lithium niobate (LiNbO_3) electro-optic waveguides [19], [20]. The structure of the IOC's phase modulation section is shown in Fig. 2. LiNbO_3 -based IOCs offer significant advantages, including a large modulation bandwidth, excellent modulation linearity, and miniaturization, with modulation bandwidths reaching tens of GHz. Phase modulation primarily exploits the linear electro-optic effect (Pockels effect) in LiNbO_3 crystals. This effect describes the phenomenon whereby the crystal's refractive index change is proportional to the applied electric field.

The half-wave voltage V_π characterizes its phase modulation capability and can be expressed as:

$$V_\pi = \frac{\pi}{K_p}, \quad (1)$$

TABLE I
TEMPERATURE SENSITIVITY COEFFICIENTS

Parameter	Temperature Sensitivity Coefficients
$dG_p/G_p dT$	$\sim 2.0 \times 10^{-6} \text{ K}^{-1}$
$dL_0/L_0 dT$	$\sim 3.2 \times 10^{-6} \text{ K}^{-1}$
$d\lambda/\lambda dT$	$\sim 7.0 \times 10^{-7} \text{ K}^{-1}$
$dn_e/n_e dT$	$\sim 1.7 \times 10^{-5} \text{ K}^{-1}$
$dr_{33}/r_{33} dT$	$\sim 4.9 \times 10^{-4} \text{ K}^{-1}$

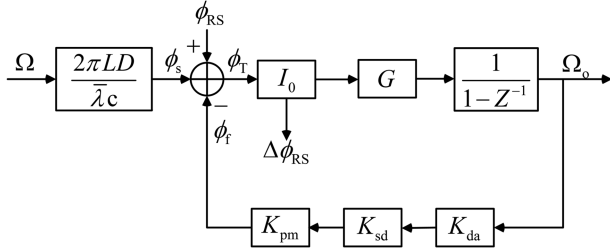


Fig. 3. Digital diagram of closed-loop IFOG system.

where K_p is the modulation coefficient, that can be expressed as:

$$K_p = \pi n_e^3 r_{33} L_0 / G_p \lambda, \quad (2)$$

where n_e is the refractive index of the waveguide, r_{33} is the electro-optic coefficient, L_0 represents the effective length of the modulator waveguide, G_p is the electrode spacing, and λ is the wavelength of the optical wave in vacuum.

In previous research, our team established a stable relationship between the half-wave voltage of lithium niobate (LiNbO₃) electro-optic modulators for IFOGs and temperature [21]. Considering the absolute typical values of the parameters in Table I, it is evident that the temperature dependence of the refractive index is the primary contributor to the IOC's temperature sensitivity. Using the typical values provided in Table I, the following can be calculated:

$$\frac{dV_\pi}{V_\pi dT} \approx -5.4 \times 10^{-4} \text{ K}^{-1}. \quad (3)$$

The digital closed-loop IFOG system is shown in Fig. 3. The total gain of the forward channel is G , K_{pm} is the phase modulation gain of the IOC, K_{sd} is the gain of the driver amplifier, and K_{da} is the gain of the DAC. Based on closed-loop feedback control principles, the ideal scale factor F_s is given by:

$$F_s = \frac{1}{K_{pm} K_{sd} K_{da}}, \quad (4)$$

where the gain of the DAC K_{da} can be calculated as:

$$K_{da} = \frac{V_\pi}{2^m}, \quad (5)$$

where m is the bit resolution of DAC. Denoting the half-wave voltage as a function of temperature T , the scale factor can be

expressed as:

$$F_s = \frac{2^m}{K_{pm} K_{sd} V_\pi(T)}. \quad (6)$$

Evidently, when the half-wave voltage V_π varies with temperature, the scale factor experiences proportional fluctuations. This directly results in significant degradation in the IFOGs scale factor performance, with temperature variations introducing scale factor errors on the order of $10^{-3} \sim 10^{-4} / ^\circ\text{C}$ [1].

B. Scale Factor Error Caused By Reset Error

The digital closed-loop system generates a feedback phase ϕ_f , which enables the servo to maintain the phase difference $\phi_T = \phi_s - \phi_f$ at the zero position. Additionally, a bias phase ϕ_b is incorporated to facilitate the operation of the IFOGs at optimal sensitivity. When $\phi_b = \pm\pi/2$, the response of the IFOG can be written by:

$$\Delta I = 2I_0 \sin \phi_T. \quad (7)$$

The step signal consists of a sequence of phase steps, each with a duration of τ and an amplitude of ϕ_f . These steps are governed by a reset mechanism, as they cannot increase or decrease indefinitely. The reset voltage V_{RS} is generated when the step signal undergoes a reset and is subsequently applied to the modulator to produce a reset phase of $\phi_{RS} = \pm 2\pi$. Then, ΔI can be expressed as:

$$\Delta I = 2I_0 \sin(\phi_T + \phi_{RS}) = 2I_0 \sin(\phi_T \pm 2\pi). \quad (8)$$

The sign of ϕ_{RS} is related to the positive and negative of the input rotation rate. In the absence of errors in ϕ_{RS} , there would be no reset error. However, this ideal situation can be easily disrupted by fluctuations in temperature.

As the gain $A = K_{pm} K_{sd} K_{da}$ of the feedback channel fluctuates with temperature, the reset phase ϕ_{RS} deviates from $\pm 2\pi$, resulting in a reset error denoted as $\Delta\phi_{RS}$. Following temperature variations, the gain of the feedback channel changes to $A' = K_{pm} K_{sd} K_{da}(1 + \varepsilon)$, leading to an affected reset phase expressed as $\phi_{RS} = \pm 2\pi(1 + \varepsilon)$. Therefore, whenever the modulation signal is reset, the reset phase error resulting from the interference is:

$$\Delta\phi_{RS} = \sin(\phi_{RS}) = \pm \sin(2\pi\varepsilon). \quad (9)$$

The step signal, input phase, and feedback phase of the IFOG are shown in Fig. 4. It is assumed that the step signal contains discrete steps ranging from the lowest level to the highest level. If the average number of steps per step signal cycle is N , the introduction of one reset phase $\Phi_{RS} = \pm 2\pi(1 + \varepsilon)$ per reset cycle, the mean equivalent input phase incorporating both the Sagnac phase and reset phase error, can be expressed as:

$$\bar{\Phi}_I = \Phi_S + \frac{\Delta\Phi_{RS}}{N} = \begin{cases} \Phi_S + \frac{\sin(2\pi\varepsilon)}{N} (\Phi_S > 0) \\ \Phi_S - \frac{\sin(2\pi\varepsilon)}{N} (\Phi_S < 0) \end{cases}. \quad (10)$$

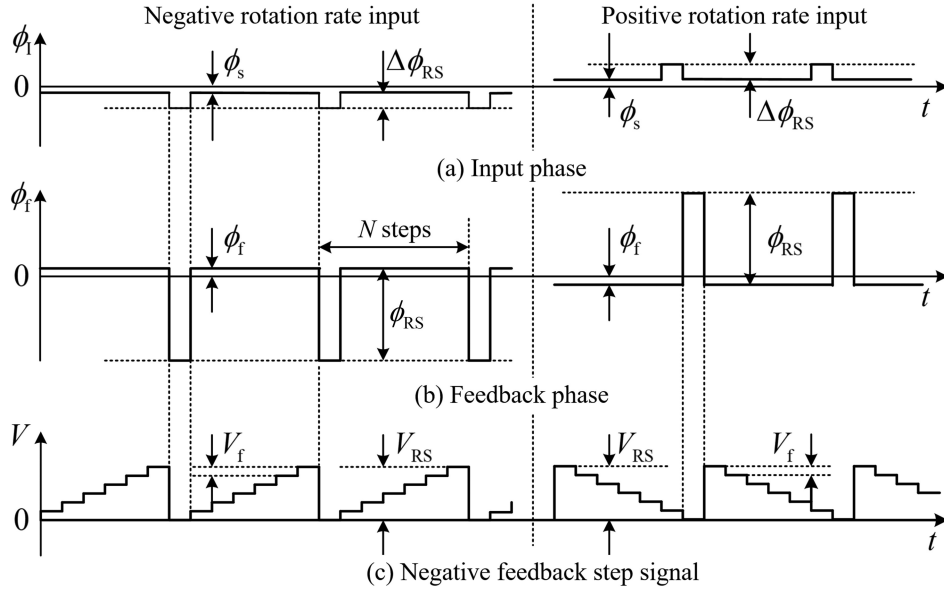


Fig. 4. The step signal, input phase, and feedback phase of IFOG.

If $\Phi_{RS} = \pm 2\pi(1 + \varepsilon)$ and the system is stable, N can be also given by:

$$N = \left| \frac{\Phi_{RS}}{\Phi_F} \right| = \left| \frac{\Phi_{RS}}{\bar{\Phi}_I} \right| = \begin{cases} \frac{2\pi(1+\varepsilon)}{\bar{\Phi}_I(\Phi_S > 0)} \\ -\frac{2\pi(1+\varepsilon)}{\bar{\Phi}_I(\Phi_S < 0)} \end{cases}. \quad (11)$$

Substituting (11) into (10) yields:

$$\bar{\Phi}_I = \frac{\Phi_S}{1 - \frac{\sin(2\pi\varepsilon)}{2\pi(1+\varepsilon)}}. \quad (12)$$

Referring to Fig. 3 and (11), the closed-loop steady-state output with reset error is:

$$\Phi'_O = \frac{\bar{\Phi}_I}{K_{pm}K_{sd}K_{da}(1+\varepsilon)} = \frac{\Phi_S}{K_{pm}K_{sd}K_{da}[(1+\varepsilon) - \frac{\sin(2\pi\varepsilon)}{2\pi}]}. \quad (13)$$

Comparing (4) and (13), the closed-loop scale factor with reset error is:

$$F'_s = \frac{1}{K_{pm}K_{sd}K_{da}[(1+\varepsilon) - \frac{\sin(2\pi\varepsilon)}{2\pi}]}. \quad (14)$$

This indicates that step signal reset error only alters the scale factor value.

When ε is substantial, the error from sine-function linear approximation becomes non-negligible. Expanding $\sin(2\pi\varepsilon)$ as a Taylor series yields:

$$\sin(2\pi\varepsilon) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{(2k-1)!} (2\pi\varepsilon)^{2k-1}. \quad (15)$$

Substituting (13) into (14):

$$F'_s = \frac{1}{K_{pm}K_{sd}K_{da}[1 - \frac{1}{2\pi} \sum_{k=2}^{\infty} \frac{(-1)^{k-1}}{(2k-1)!} (2\pi\varepsilon)^{2k-1}]}. \quad (16)$$

Equation (14) reveals that the relative variation ε of affects the scale factor only through a third-order error term in $K_{pm}K_{sd}K_{da}$.

To constrain its impact on the scale factor below 10^{-5} :

$$\left| \frac{1}{2\pi} \sum_{k=2}^{\infty} \frac{(-1)^{k-1}}{(2k-1)!} (2\pi\varepsilon)^{2k-1} \right| < 10^{-5}. \quad (17)$$

Neglecting higher-order terms in (17) yields:

$$|\varepsilon| < 0.0063. \quad (18)$$

Of course, even in a temperature-controlled environment, reset error cannot be completely eliminated. This is because the negative feedback voltage signal for the digital closed-loop IFOG originates from the DAC driving the IOC. The bit-width limitation of the DAC results in quantization of its output modulation voltage. For an ideal m -bit DAC, through digital signal rounding, we can constrain its quantization error to within half of the Least Significant Bit (LSB), i.e.,:

$$\varepsilon_{da} = \frac{1}{2^m}. \quad (19)$$

If only the impact of DAC quantization on reset error is considered, then to maintain scale factor error below 10^{-5} , the condition $1/2^m < 0.0063$ must be satisfied, which requires $m \geq 8$.

III. COMPENSATION OF HALF-WAVE VOLTAGE THERMAL DEPENDENCE ERRORS ON SCALE FACTOR

A. The Principle and Limitations of Secondary Feedback Loop

To mitigate performance degradation in IFOGs caused by temperature-induced variations in the IOC's half-wave voltage V_π , a secondary feedback loop has been developed and widely implemented. Under ideal conditions with the rotation rate detection loop operating normally, if V_π remains unchanged before and after reset, (8) indicates that the detector-received interference intensity should be identical during these reset phases. However, when V_π shifts between reset cycles, a measurable

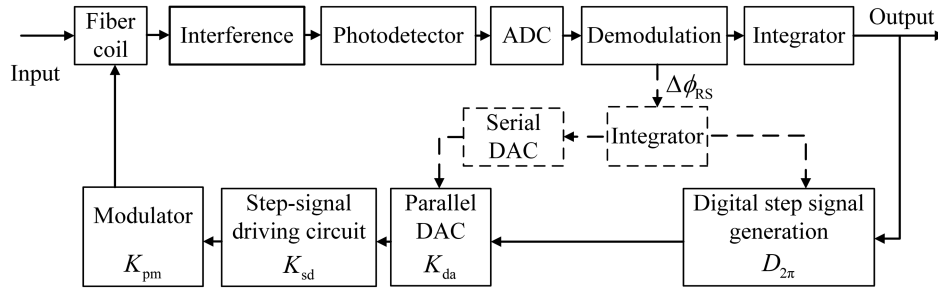


Fig. 5. Diagram of digital closed-loop IFOG with secondary feedback loop control.

discrepancy in interference intensity occurs. The secondary feedback loop compensates by measuring the difference in demodulated angular velocity values between adjacent half-cycles of modulated signal during reset events. Through servo-controlled adjustment of the modulation coefficient, this error has been greatly reduced, achieving effective V_π error compensation.

The diagram of the digital closed-loop IFOG implementing secondary feedback loop is shown in Fig. 5. The dashed-line section represents the secondary feedback loop. When the gain $K_{pm}K_{sd}K_{da}$ deviates from its ideal value, analysis indicates that:

$$\Phi_{RS} = K_{pm}K_{sd}K_{da}D_{2\pi} \neq 2\pi. \quad (20)$$

Evidently, adjusting parameters K_{pm} , K_{da} , K_{sd} , or $D_{2\pi}$ through secondary feedback loop maintains the reset phase at 2π . Among these, parameters K_{da} and $D_{2\pi}$ offer convenient adjustment: Parameter K_{da} can be regulated by adjusting the reference voltage of the parallel DAC through a serial DAC, while parameter $D_{2\pi}$, being a digital quantity, can be adjusted by configuring the reset register within the FPGA. The demodulated reset error $\Delta\Phi_{RS}$ generates a control signal, which passes through an integration stage and proportional stage to adjust either parameter K_{da} or $D_{2\pi}$.

However, it must be recognized that this method is not universally applicable due to potential instability in its closed-loop operation. The secondary feedback loop's functionality depends on rotation rate demodulation values during resets, while the reset frequency of the step signal varies with input rotation rates. This relationship originates from closed-loop feedback principles: the step height of the step signal corresponds precisely to the rotation rate detected by the gyroscope. Under low rotation rate inputs, step signal resets occur infrequently. Conversely, high rotation rate inputs trigger frequent resets. For a low-precision IFOG with a bias stability of $0.1^\circ/\text{h}$, the dynamic range typically extends from hundredths of degrees per hour to hundreds of degrees per second—a variation exceeding 10^7 in magnitude. Consequently, the reset interval varies by a factor of 10^7 across this operational range.

B. Modeling of Half-Wave Voltage Temperature Dependence

Building upon the well-established linear relationship between the IOC half-wave voltage and temperature, effective

compensation can be achieved using a simple linear regression model. Implementation requires only basic multiply-accumulate operations within the FOG's feedback control software. Real-time temperature monitoring is facilitated by integrating a digital temperature sensor at the waveguide, introducing negligible hardware overhead. Temperature-induced scale factor effects are compensated through FPGA-based adjustment of digital modulation parameters. Fig. 6 illustrates the IFOG schematic incorporating the IOC half-wave voltage temperature dependence compensation model, with the implementation steps detailed below.

- 1) The half-wave voltage V_π of the IOC should be measured under various temperature conditions, typically within the operational temperature range of IFOGs. Numerous studies have documented the methodologies for measuring the V_π , and these methods are relatively straightforward to implement. Currently, automated measurement equipment for the V_π is commercially available, rendering it suitable for batch measurements. The relationship between V_π and ambient temperature should be modeled as $V_\pi(T)$ through a fitting process. Typically, a first-order polynomial fit is adequate to achieve satisfactory accuracy. Establish the $V_\pi(T)$ coefficient and temperature relationship model:

$$c(T) = C(V_\pi(T)). \quad (21)$$

- 2) Establish the closed-loop feedback gain and compensation model:

$$A = K_{pm}K_{sd}K_{da}(1 + \varepsilon) \cdot c_j(\bar{T}_j) = A' \cdot c_j(\bar{T}_j), \quad (22)$$

and implemented within the FPGA. During the operation of the IFOG, temperature data collected internally can be input into the model, which subsequently adjusts the feedback gain to achieve real-time compensation for half-wave voltage.

IV. VALIDATION TESTING FOR COMPENSATION METHOD

A. Half-Wave Voltage Temperature Relationship Model

To establish the relationship between the IOC's half-wave voltage V_π and ambient temperature, V_π was measured at various temperatures using a four-step phase modulation-based testing method as Fig. 7. A single-axis gyroscope was assembled using the aforementioned IOC, a fiber coil, a light source, a coupler, a detector, and detection circuitry. The temperature sensor was

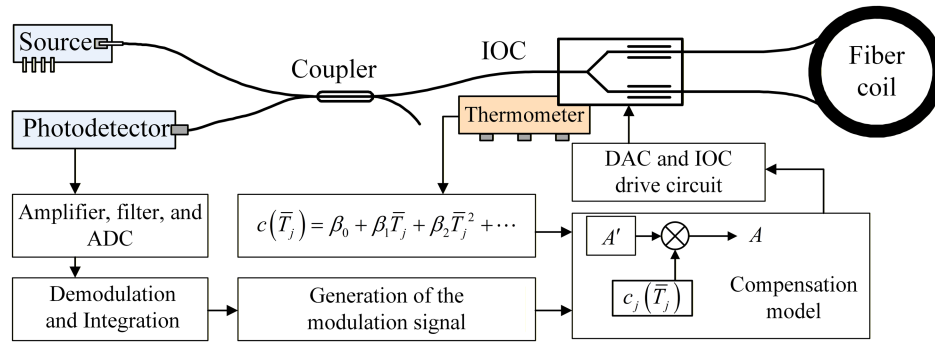


Fig. 6. Schematic diagram of IFOG with half-wave voltage temperature dependence error compensation model.

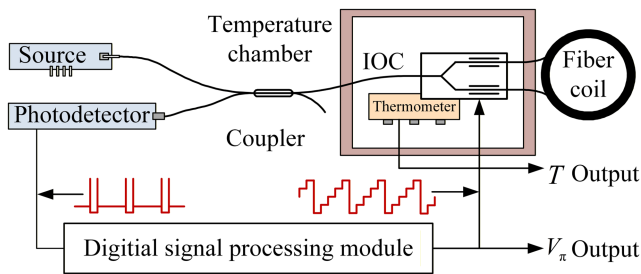


Fig. 7. IOC half-wave voltage measurement structure using the Four-step phase modulation method.

securely affixed adjacent to the IOC. The optical path operates at a wavelength of $1.3 \mu\text{m}$, which is commonly used in medium-to-low precision IFOGs. The source is a superluminescent diode (SLD) equipped with a thermoelectric cooler (TEC), exhibiting a spectral bandwidth of approximately 20 nm. A single-mode coupler with a 50:50 splitting ratio is employed, and the detector has a measurement bandwidth exceeding 80 MHz. The digital signal processing system automatically adjusts the amplitude of the staircase waveform signal to equalize the two interference intensity values before and after the staircase reset. This enables efficient and accurate calculation of the IOC's V_π under current ambient conditions, with errors below 100 μV .

The IOC under test with an operating wavelength range of $1.3 \mu\text{m} \pm 0.05 \mu\text{m}$, modulation bandwidth $>200 \text{ MHz}$, single-mode polarization-maintaining fiber pigtailed, operating temperature range from -40°C to 85°C . An integrated digital temperature sensor measures waveguide temperature with accuracy better than 0.5°C , measurement range covering the IOC's full operating temperature range, and compact size for easy installation. A programmable thermal chamber controls the IOC's ambient temperature with control accuracy better than 1°C . The measurement system is depicted in the accompanying Fig. 8.

The temperature sensor was affixed to the IOC, and the IOC was placed in the thermal chamber. The chamber temperature was programmed to increase from -20°C to 50°C at a rate of $0.5^\circ\text{C}/\text{min}$. V_π measurements were taken at 1°C intervals, the test results are shown in Fig. 9. Test results exhibit non-linearity

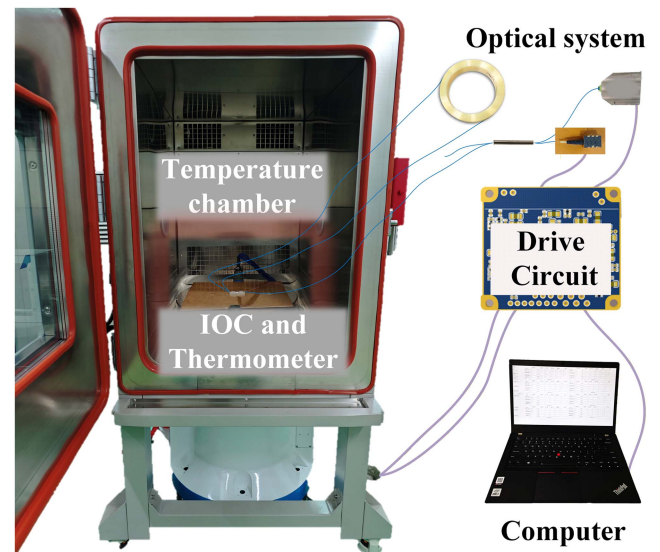


Fig. 8. IOC Half-wave voltage measurement system.

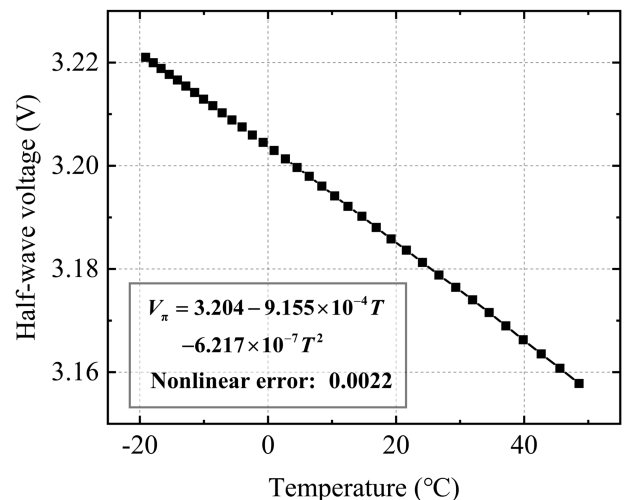


Fig. 9. Correlation between half-wave voltage and temperature.

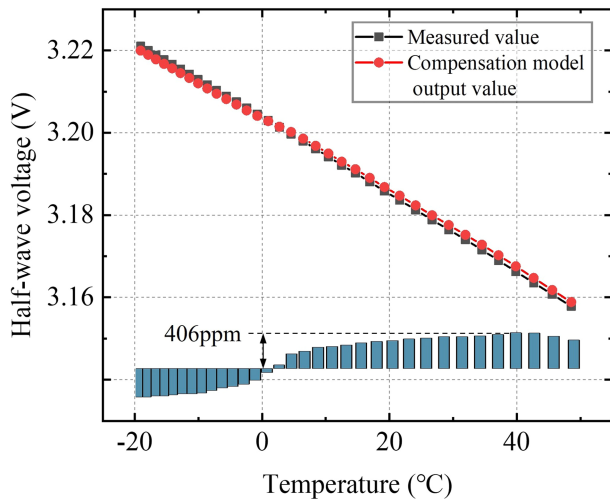


Fig. 10. Half-wave voltage compensation test results.

below 0.22%, further validating the stable linear relationship between the IOC's V_π and ambient temperature. Based on these measurements, linear regression was applied to fit the data. When a second-order fit is employed, it is evident that coefficient -6.217×10^{-7} has a negligible effect on the variation of the half-wave voltage. Consequently, a reset error compensation model for the modulation signal is established as:

$$V_\pi = 3.204 - 9.315 \times 10^{-4}T. \quad (23)$$

B. Modeling Verification

The validation experiment continued using the experimental setup depicted in Fig. 8. The compensation model was programmed into the FPGA of the signal processing circuit, with model-generated compensation coefficients output via serial communication. The chamber temperature was programmed to ramp from -20°C to 50°C at $0.5^\circ\text{C}/\text{min}$. V_π measurements were taken at 1°C intervals, the test results are shown in Fig. 10. The maximum and average differences between the voltage values estimated by the compensation model and the actual measurements are 406 ppm and 277 ppm, respectively. According to the previous analysis, this level of tracking accuracy is sufficient to reduce the scale factor error induced by IOC temperature variations to an extent that is acceptable for most medium-to-low precision IFOG applications. This result confirms that the compensation model effectively tracks variations in the half-wave voltage.

V. CONCLUSION

This paper analyzes the impact of the thermal dependence of the half-wave voltage in IOCs for IFOGs on scale factor performance. It also examines how reset errors in feedback staircase signals (generated when modulation voltage deviates from half-wave voltage during resets) affect scale factor accuracy, establishing a quantitative relationship between reset errors and scale factor inaccuracies. Analysis indicates that modulation

signal reset errors do not affect scale factor asymmetry or nonlinearity. Specifically, it is determined that when reset errors are below 0.0063, the resulting scale factor error can be maintained within 10 ppm. The proposed analysis aids in optimizing IFOG design, providing references for selecting more effective and cost-efficient strategies to mitigate reset errors.

Furthermore, this paper clarifies the operational principles and limitations of the secondary feedback loop for compensating IOC half-wave voltage errors. Based on the stable relationship between IOC half-wave voltage and temperature, an economical and efficient compensation method for temperature-induced half-wave voltage errors is proposed, with detailed implementation steps provided. Experimental results demonstrate that this method achieves a compensation accuracy of 406 ppm for half-wave voltage. Compared to the secondary feedback loop, this approach is simpler to implement and offers a compensation accuracy. Therefore, it is particularly well-suited for low to medium precision measurement applications and time-division multiplexing interferometric IFOGs.

Given that LiNbO_3 modulators are not only key components in IFOGs but also critical for various optical sensors requiring modulation—such as optical fiber strain sensors, optical current sensors, and fiber-optic hydrophones—the impact of thermal drift in LiNbO_3 devices on measurement accuracy is a widely recognized concern. Therefore, both the analysis and the proposed method in this study hold general relevance for high-precision sensors employing LiNbO_3 technology.

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